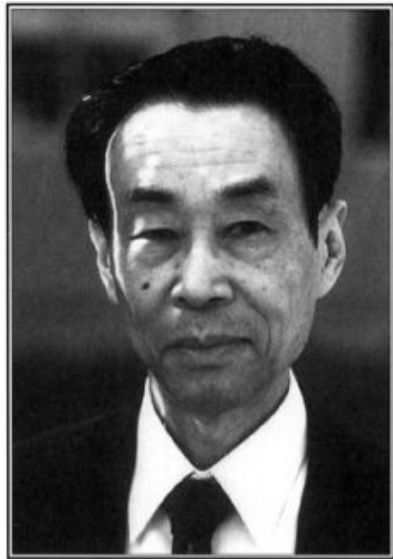


Neutral Theory of Molecular Evolution

- 1. most mutations are harmful and thus removed by “negative” (or “purifying”) natural selection.**
- 2. some mutations are neutral and thus accumulate in natural populations by random genetic drift.**
- 3. very rarely, beneficial mutations occur and are fixed by “positive” Natural selection.**
- 4. The rate of evolution of a molecule is determined by its degree of “functional constraint”.**

Neutral Theory of Molecular Evolution



MOTOO KIMURA

- 1) For each protein, the rate of evolution (amino acid substitution) is approximately constant per generation as long as the functional constraint (f_o , fraction of neutral mutations) remains constant
- 2) Functionally less important molecules or parts thereof will have a higher substitution rate than functionally important ones
- 3) Those substitutions that disrupt proteins less will occur more frequently than those that disrupt more
- 4) Selective elimination of alleles and the random fixation of neutral alleles are more prevalent than alleles fixed by positive Darwinian selection
- 5) Polymorphism is a transient phase of molecular evolution (substitution), not an adaptive stable state. Constant input of mutations, not natural selection, is the cause of polymorphism.

Different molecular regions, different rates

- DNA distant from genes evolves very quickly (at about one substitution per 10^8 years),
- Flanking regions upstream and downstream from a gene evolve less quickly than that,
- Introns evolve less quickly than those, though not much less,
- Third positions of codons evolve less quickly than introns,
- First and second positions of codons evolve less quickly than that,

Functionally Important Parts of the Gene Sequence Evolve More Slowly Than Other Parts

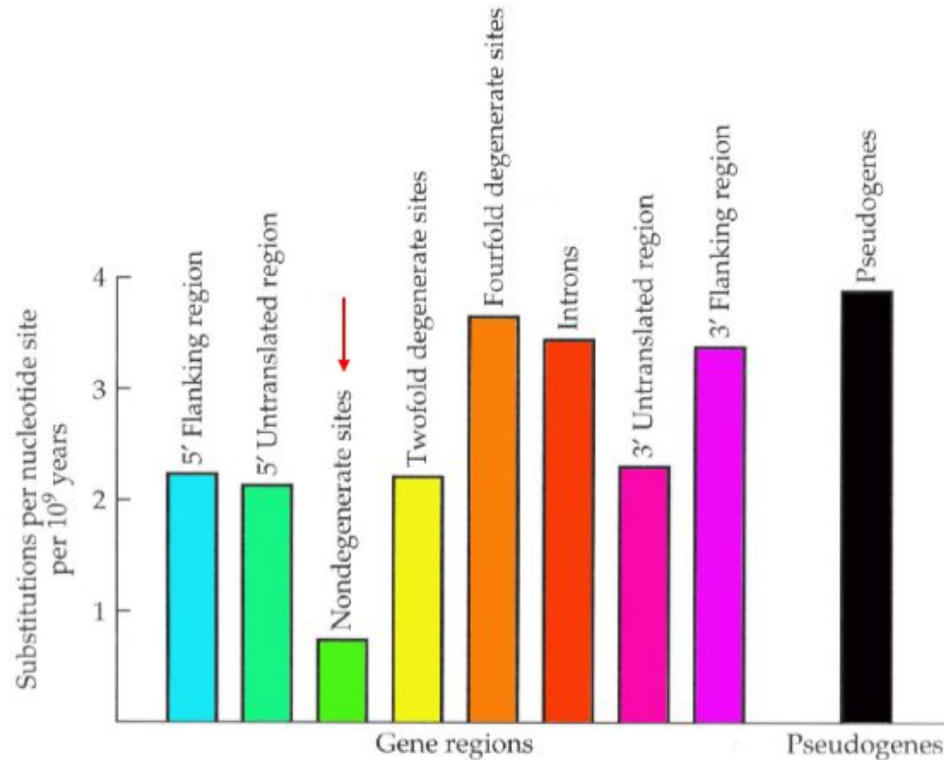
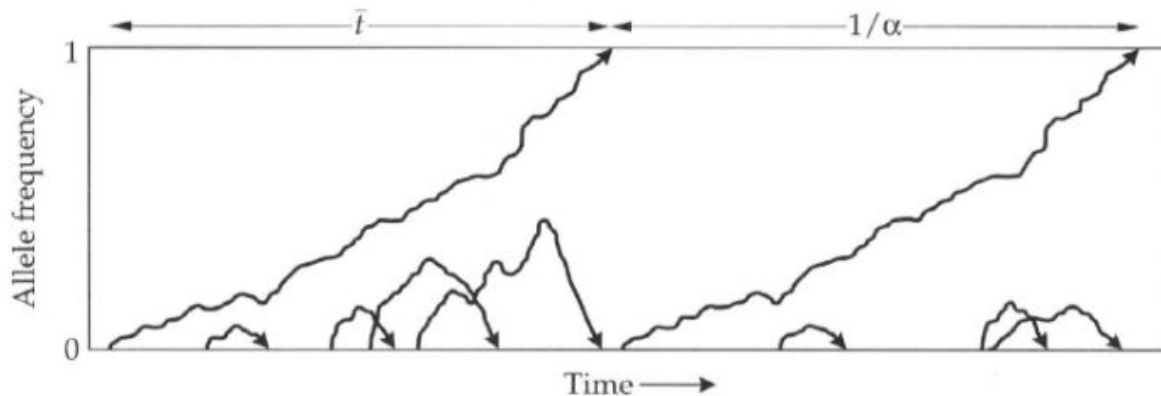


Figure 7.2 Average rates of substitution in different parts of genes and in pseudogenes. After Li and Graur (1991).

The Fixation probability of Neutral Mutations



- For any locus, $2N\mu$ mutations enter a population each generation.
- Each new neutral mutation has a probability of being fixed equal to its initial frequency, $1/(2N)$.
- The average number of substitutions per locus per generation is equal to the product of these,

$$2N\mu \times 1/(2N) = \mu$$

Major Shocker from the Neutral Theory

- The rate of substitution equals the rate of mutation (when there is no selection).
- Evolution should be nearly constant because rate of mutation is nearly constant.
- Since rates of substitution is independent of population size, small or large populations evolve at a similar rate.

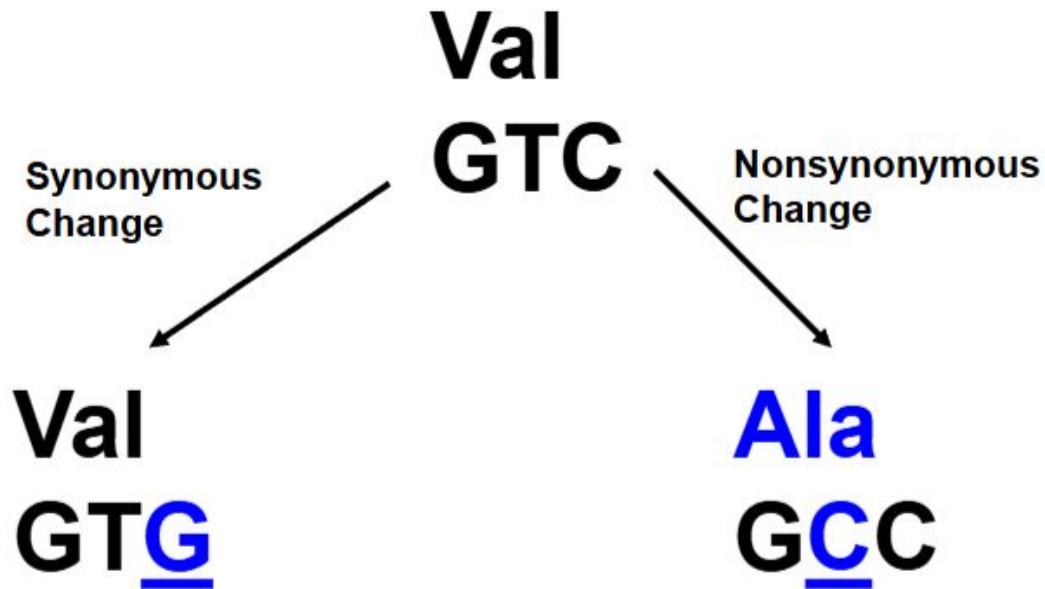
Neutral Theory-O-Rama

- 1) Population size does not matter
 - drift stronger in small populations
 - *but* mutation equally less common!
- 2) Selection for new beneficial genes does not matter
 - most new mutations harmful, eliminated by natural selection.
 - v therefore approximates the maximum rate of evolutionary change.

2 types of changes in codons

Synonymous = silent change (amino acid stays the same)

Nonsynonymous = replacement change (changes amino acid)



d_N = # nonsynonymous substitutions/# nonsynonymous sites

d_S = # synonymous substitutions/# synonymous sites

Test for selection by comparing d_N and d_S

$d_N/d_S = 1$: Neutral evolution

$d_N/d_S < 1$: Purifying selection

$d_N/d_S > 1$: Positive selection

The d_N/d_S ratio (ω) measures the selective pressure

Example:

Degeneracy	1	112	114	114	114
		Asp	Thr	Ala	Val
Sequence	1	GAC	ACA	GCG	GTT

How many nonsynonymous sites in sequence 1?
8 nondegenerate sites 1 two fold degenerate site
= 8.66 nonsynonymous sites

Example:

Degeneracy 1	112	114	114	114
	Asp	Thr	Ala	Val
Sequence 1	GAC	ACA	GCG	GTT
Sequence 2	GCC	ACT	TCG	GTT
	Ala	Thr	Ser	Val
Degeneracy 2	114	114	114	114

Sequence 2 has 8 nonsynonymous sites and 4 synonymous sites.

For this comparison, we average number from both sequences.

Nonsynonymous sites = $(8.66 + 8)/2 = 8.33$

Synonymous sites = $(3.33 + 4) = 3.67$

Example:

Degeneracy 1	112	114	114	114
	Asp	Thr	Ala	Val
Sequence 1	GAC	ACA	GCG	GTT
Sequence 2	GCC	ACT	TCG	GTT
	Ala	Thr	Ser	Val
Degeneracy 2	114	114	114	114

There are 2 nonsynonymous changes,

So $dn = 2/8.33 = 0.24$

There is 1 silent change,

So $ds = 1/3.67 = 0.27$

$dn/ds = 0.23/0.27 = 0.88$

< 1 despite having more nonsynonymous changes.

Predictions of the neutral theory

1. The level of within species genetic variation is determined by population size and mutation rate, and is correlated with the level of sequence divergence between species.
2. The rate of gene evolution (substitution) is inversely related to the level of functional constraint (purifying selection) acting on the gene. [this is macro-evolution]
3. The pattern of base composition at neutral sites reflects mutational equilibrium.
4. There is a constant rate of sequence evolution; i.e., a molecular clock.

Neutral theory is “the” rigid null hypothesis for molecular evolution

Misconceptions about neutral theory

There has been some confusion about what the neutral theory suggests, so it is worth trying to clear up some of the misconceptions.

Myth 1: Only genes that are unimportant can undergo neutral mutations. [Neutral theory only asserts that alternative alleles segregating at a locus are selectively equivalent. Such loci can, and do, encode genes that have important functional roles.]

Myth 2: Neutral theory diminished the role of natural selection in adaptation. [To the contrary, neutralists and selectionists both maintain that natural selection is the primary mechanism of adaptation, and that morphological evolution is primarily driven by natural selection.]

Myth 3: Nucleotide or amino acid sites that undergo neutral substitutions are not subject to natural selection. [Neutral theory does not preclude the possibility that adaptive mutations can occur at sites where neutral mutations occur. Neutral theory only asserts that adaptive mutations will be much less frequent and will go to fixation much more quickly; hence, most polymorphism observed in a population will be neutral.]

Myth 4: Neutral mutations have a selective coefficient of $s = 0$. [Because natural population sizes are finite, the fate of mildly deleterious alleles can be fixed due to drift..]

Myth 5: Neutral mutations are always neutral. [Neutral theory makes no assertions about the stability of the environment. The selection coefficient will depend of the environment.]

- Migration = **Gene Flow**

- the movement of alleles between putative populations.
- gene flow binds biological species together.

- Genetic Drift = **Sampling Error**

- an inevitable feature of finite populations.
- Both forces are agents of evolutionary change.



Genetic Drift: Sampling error during meiosis

- Another opportunity for chance to influence the fate of genes comes during meiosis.
- When an individual is heterozygous at a locus, only one of the two alleles is passed to each gamete that it makes.
- If you have two children, there is a chance of $1/2$ that they will both inherit the same allele from you, and your other allele will leave no descendants in the next generation.
- So even if all individuals survive and leave the same number of offspring, meiosis itself causes random changes allele frequencies.

Drift is unbiased

random fluctuations in allele frequency are larger in smaller populations

drift causes genetic variation to be lost.

drift causes populations that are initially identical to become different

an allele can become fixed without the benefit of natural selection